

Decision Log – Aerial GCP Pose Estimation

Project Objective

The objective of this project was to develop a machine learning system capable of performing two tasks simultaneously from aerial survey imagery:

1. Ground Control Point (GCP) Localization
2. Predict the precise center coordinates of the marker.
3. Marker Shape Classification
4. Predict whether the marker belongs to one of the following classes:
 - Cross
 - Square
 - L-Shape

The assignment intentionally provided limited guidance regarding architecture selection, data handling strategy, and inference methodology. Therefore several engineering decisions had to be made based on dataset analysis and practical considerations.

Dataset Understanding

Before designing the model, a detailed Exploratory Data Analysis (EDA) was performed.

Dataset Statistics

Metric	Value
Training Images	1000
Test Images	300
Projects	11
Surveys	14
Physical GCP Groups	159

The dataset follows a nested production-style structure:

project_name/ └── survey_name/ └── gcp_id/ └── image.JPG

Unlike many academic datasets, multiple images can belong to the same physical Ground Control Point.

Shape Distribution

Shape	Count
L-Shape	491
Square	328
Cross	177
Missing Labels	4

Observations

- The dataset is moderately imbalanced.
 - Cross markers are significantly less frequent.
 - Four samples contain coordinate annotations but lack shape labels.
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Resolution Analysis

The dataset contains images with resolutions approximately:

- 4096×3068
- 4096×2730

Observations

The marker occupies only a tiny fraction of the image.

Directly predicting coordinates from the full image would require the network to localize a very small object among millions of pixels.

Coordinate Distribution Analysis

Marker coordinates span nearly the entire image area.

Observations

- No center bias exists.
- Markers appear close to borders as well as near image centers.

This ruled out any assumption that markers would always appear near the center of the image.

Dataset Correlation Analysis

A significant characteristic of the dataset is that multiple images belong to the same physical GCP.

Images within the same GCP folder often share:

- Similar viewpoints
- Similar terrain
- Similar lighting conditions
- The same marker

This observation strongly influenced the validation strategy.

Engineering Decisions

Decision 1: Use GroupKFold Instead of Random Splitting

Observation

Multiple images belonged to the same physical GCP.

Problem

A random image-level train-validation split could place highly correlated images in both training and validation sets.

This would artificially inflate validation performance.

Alternatives Considered

1. Random train-validation split
2. Stratified split
3. GroupKFold

Chosen Solution

GroupKFold

Grouping key:

project/survey/gcp_id

Reasoning

This guarantees that images from the same physical marker never appear in both training and validation sets.

Expected Impact

- Leakage-free validation
 - More realistic performance estimation
 - Better generalization measurement
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Decision 2: Reformulate the Problem Using Crops

Observation

The original images are approximately 4096×3000 pixels.

The marker occupies only a tiny portion of the image.

Problem

Direct full-image coordinate regression is difficult and computationally inefficient.

Alternatives Considered

1. Full-image coordinate regression
2. Object detection pipeline
3. Crop-based localization

Chosen Solution

Generate marker-containing crops during training.

Reasoning

The crop-based formulation transforms a global localization problem into a local coordinate regression problem.

This significantly simplifies learning.

Expected Impact

- Faster convergence
 - Reduced memory requirements
 - Better localization accuracy
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Decision 3: Use EfficientNet-B3 Backbone

Observation

The dataset contains only 1000 labeled images.

Problem

Training a large model from scratch would likely overfit.

Alternatives Considered

1. ResNet50
2. EfficientNet-B3
3. Vision Transformer

Chosen Solution

ImageNet-pretrained EfficientNet-B3

Reasoning

EfficientNet-B3 provides:

- Strong transfer learning performance
- Excellent accuracy-to-parameter ratio
- Lower computational requirements

Expected Impact

- Improved generalization
 - Faster convergence
 - Reduced training time
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Decision 4: Use Multi-Task Learning

Observation

Localization and shape classification depend on the same visual marker features.

Problem

Training separate models would duplicate feature extraction.

Chosen Solution

Shared backbone with:

- Regression head
- Classification head

Reasoning

Shared features improve efficiency while allowing both tasks to learn simultaneously.

Expected Impact

- Reduced model complexity
 - Better feature utilization
 - Faster inference
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Decision 5: Ignore Missing Shape Labels

Observation

Four samples contain coordinates but lack shape labels.

Problem

Discarding those samples wastes localization information.

Chosen Solution

Ignore classification loss using:

```
ignore_index = -100
```

while continuing regression training.

Reasoning

Coordinate supervision remains valuable even without shape annotations.

Expected Impact

- Better data utilization
 - Improved regression learning
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Decision 6: Use Tile-Based Inference

Observation

The test dataset provides only images.

No marker coordinates are available.

Problem

The model must identify marker locations in unseen images.

Alternatives Considered

1. Whole-image inference
2. Sliding-window inference

Chosen Solution

Tile-based inference

Reasoning

The image is divided into overlapping tiles.

The tile with the highest classification confidence is selected.

Expected Impact

- Better localization performance
- Improved robustness on large images

Decision 7: Use Macro F1 Score

Observation

Class imbalance exists.

Problem

Accuracy alone can be misleading.

Chosen Solution

Track Macro F1 Score in addition to accuracy.

Reasoning

Macro F1 evaluates all classes equally regardless of frequency.

Expected Impact

More reliable classification evaluation.

Decision 8: Two-Stage Training Strategy

Chosen Solution

Stage 1:

- Freeze backbone
- Train task-specific heads

Stage 2:

- Unfreeze full network
- Fine-tune end-to-end

Reasoning

This allows the model to adapt gradually while preserving useful pretrained features.

Expected Impact

- Improved stability
 - Better convergence
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Rejected Approaches

Rejected Approach 1: Random Train-Validation Split

Rejected because EDA revealed strong correlations between images belonging to the same GCP.

This would create severe data leakage.

Rejected Approach 2: Full-Resolution Training

Rejected because:

- High memory requirements
- Small target size
- Slower training

Crop-based training provided a more practical solution.

Rejected Approach 3: Separate Models for Each Task

Rejected because both tasks depend on similar visual features.

A shared backbone was more efficient.

Lessons Learned

The most important lesson from this project was that understanding the dataset had a greater impact than choosing a complex architecture.

The largest improvements came from:

- Leakage prevention
- Crop-based training
- Proper validation design
- Robust inference methodology

This reinforces the importance of data-centric machine learning for real-world computer vision systems.

Limitations

1. The model assumes exactly one marker per image.
 2. Test-set quantitative evaluation is not possible because labels are unavailable.
 3. Some challenging samples still produce larger localization errors.
 4. Tile selection relies on classification confidence as a heuristic.
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Future Work

Potential improvements include:

1. Heatmap-based keypoint localization.
 2. Higher-resolution crop training.
 3. Cross-fold ensembling.
 4. Confidence calibration.
 5. Vision Transformer backbones.
 6. Multi-scale tile inference.
 7. Full 5-fold model averaging.
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Final Outcome

The final system successfully performs:

- Ground Control Point localization
- Marker shape classification
- Test-set inference

A complete predictions.json file was generated for all 300 test images.

The final solution remains modular, reproducible, and aligned with the engineering constraints identified during dataset analysis.